

Curriculum Vitae/Resume

Seunghun Oh

✉ alvin0808@snu.ac.kr 🔗 alvin0808.github.io 📄 Google Scholar

Education

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- University of California, Los Angeles (UCLA)** Starting Sep 2026
*Incoming PhD Student in **Civil and Environmental Engineering***
- Seoul National University (SNU)** Mar 2020 - Aug 2026
*BS in **Mechanical Engineering**, Double Major in **Computer Science***
- **GPA: 3.94/4.0, Major GPA: 3.97/4.0** 5th of 92, Mechanical Engineering, SNU
 - 21 months of military service included
- Gyeonggi Science High School for the Gifted (GSHS)** Mar 2017 - Feb 2020
*Major in **Physics and Mathematics***
- 1 Year Early Entrance

Research Interests

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- Trustworthy robot learning and mobile autonomy for navigation, autonomous driving, and human-centered robotics
 - Robot perception and 3D spatial intelligence for SLAM, reconstruction, and multi-sensor fusion

Publications and Preprints

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- D. Zhu, **S. Oh**, R. Huang, Z. Huang, J. Ma, and C. Tang*
 “SCAPE: Scenario-Conditioned Simulation-Augmented Policy Evaluation”
Conference on Robot Learning (CoRL), 2026. [arXiv:2608.19425](#) 🔗
 - **S. Oh**, J. Choe, D. Lee, D. Lee, S. Jeong, Y.-C. F. Wang, and J. Park*
 “SVRecon: Sparse Voxel Rasterization for Surface Reconstruction”
arXiv preprint. arXiv:2511.17364 🔗, 2025. [\[Project Page\]](#) 🔗
 - H. Song, D. Lee, **S. Oh**, M. Jung, and A. Kim*
 “The City That Never Settles: Simulation-based LiDAR Dataset for Long-term Place Recognition under Extreme Structural Changes”
IEEE International Conference on Robotics and Automation (ICRA) Workshop, 2025.
Best Research Award. [arXiv:2505.05076](#) 🔗
 - **S. Oh**[†], Y. Kim[†], C. Song, and A. Kim*
 “LiDAR Data Processing Algorithm for Robust 6-DoF Estimation Using Circular Patterns”
Journal of the Korean Robotics Society (KROS), June 2025. [KROS Journal Link](#) 🔗
 - S. Hahn[†], **S. Oh**[†], M. Jung, A. Kim, and S. Jung*
 “Quantitative 3D Map Accuracy Evaluation Hardware and Algorithm for LiDAR (-Inertial) SLAM”
International Conference on Control, Automation and Systems (ICCAS), 2024. [arXiv:2408.09727](#) 🔗

† Co-first authors. * Corresponding author.

Patents

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- D. Lee, J. Lee, W. Kim, **S. Oh**, and J. Park*
 “Method and Apparatus for 3D Reconstruction and Articulated Structure Estimation from Rest-State Multi-View Images” *Korean patent application in preparation.*

Research Experiences

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- Incoming Ph.D. Student**, UCLA TAMS Lab (Advisor: Prof. Chen Tang) Los Angeles, CA, USA
University of California, Los Angeles Feb 2026 – Present
- **Researching inference-time safety guidance for diffusion policies**
 Developing residual-reward guidance methods that adapt diffusion-policy action chunks at inference time while preserving the original task behavior.
 - **Developed SCAPE for scenario-conditioned policy evaluation**
 Built a simulation-augmented evaluation framework that combines limited real-world rollouts with large-scale simulation to predict scenario-specific robot policy performance.
- Undergraduate Researcher**, SNU-VGI Lab (Advisor: Prof. Jaesik Park) Seoul, Korea
Seoul National University Sep 2025 – Feb 2026
- **Developed a novel “Rest-State” framework for Articulated 3D Reconstruction**
 Contributed ideas to the core pipeline design and ran baseline experiments on large-scale scenes.

- **Advanced SDF-based Sparse Voxel Rasterization with 3D Foundation Models**
Developed *SVRecon*, an SDF-based sparse voxel rasterization framework using Pi3-based initialization and hierarchical smoothness losses, achieving cleaner surfaces and faster convergence than SVRaster.

Undergraduate Researcher, RPM Robotics Lab (Advisor: Prof. Ayoung Kim)

Seoul, Korea

Seoul National University

Dec 2023 – Aug 2025

- **3D Reconstruction with SDF-based Sparse Voxel Rasterization**
Developed an SDF-based voxel rasterization pipeline to enhance mesh accuracy, extending SVRaster’s architecture. Focused on enforcing voxel-wise continuity using CUDA and PyTorch. → [Open-source project](#) 
- **Long-term Place Recognition using Simulated LiDAR Datasets**
Adapted and implemented long-term loop closure algorithms (Scan Context, BTC, Solid) in simulated urban environments. Evaluated PR performance under structural changes with customized pipelines.
- **LiDAR-Camera Calibration via Circular Marker Detection**
Developed a robust LiDAR-based 6-DoF pose estimator using circular patterns, later integrated with *DiscoCal (CVPR)* into a full LiDAR-camera calibration system. → [Open-source project](#) 
- **Quantitative Evaluation of 3D SLAM Map Accuracy**
Designed experimental setups and metrics for evaluating LiDAR(-inertial) SLAM accuracy. Developed benchmarking tools for 3D map reconstruction quality comparison. → [Open-source project](#) 

Research Student, Aerospace Engineering Club (Bulnabi)

Seoul, Korea

Seoul National University

Dec 2023 – Apr 2024

- Built foundational robotics environment using Ubuntu and ROS
- Simulated drone operations with Gazebo and QGroundControl

Research Student, Mechanical Engineering Club (ZERO)

Seoul, Korea

Seoul National University

Mar 2020 – Aug 2020

- Developed low-level C/C++ control modules in a Linux-based robotics environment.

Selected Awards and Honors

Scholarships / Fellowships

- **Kwanjeong Overseas Scholarship**, Kwanjeong Educational Foundation Fall 2026 – Spring 2030
Annual scholarship of \$25,000–\$30,000 supporting Ph.D. research at UCLA.
- **The National Presidential Science Scholarship**, Korea Student Aid Foundation 2024 – 2025
Full tuition plus approx. \$3,700/year stipend for **four semesters**, awarded by the Korean government to top STEM undergraduates.
- **Work-Study Scholarship** (*Value Exploration and Practice*), Seoul National University Mar 2025 – Nov 2025
Work-study research scholarship supporting an independent undergraduate project (approx. \$1,500).
- **Work-Study Scholarship** (Type 1), Seoul National University Mar 2024 – Nov 2024
Work-study research scholarship supporting an independent undergraduate project (approx. \$2,200).
- **Merit-based Scholarship**, Dept. of Mechanical Engineering, Seoul National University Spring – Fall 2021

Awards and Honors

- 2026 – **IPIU Outstanding Paper Award**, Encouragement Prize, for “Sparse Voxel Rasterization for 3D Surface Reconstruction from Images” (selected for **oral presentation**)
- 2025 – Co-authored ICRA 2025 workshop paper [1]; awarded **Best Research Award** (1st Place), IEEE ICRA Workshop on Future of Construction
- 2025 – **Outstanding Undergraduate Research Opportunities Program Award**, Dept. of Mechanical Engineering, Seoul National University, selected **1 of 3** awardees from a **57-student** cohort (top ~5%)
- 2024 – **Grand Prize**, Mechatronics Competition
- 2022 – **Winner**, Startup Camp hosted by Hanbat University (During Military Service)
- 2021 – Academic Honor: Highest GPA in 2nd Semester (4.3/4.3)
- 2018 – **Training Lineup for IPhO (International Physics Olympiad)**
- 2018 – **Winner**, SNU Youth Engineering Frontier Camp

Presentations / Talks

- Poster Presentation, “LiDAR Data Processing Algorithm for Robust 6-DoF Estimation Using Circular Patterns”
Presented at KROS Spring Conference, Gangwon, Korea, 2025
- Seminar Talks, “Introduction to Linux”, “Drone software & simulation” “Introduction to ROS” [link](#) 
Robotics Club Seminar, Seoul National University, 2024

Leadership / Extracurricular Activities

Republic of Korea Air Force (ROKAF)

Sergeant, 15th Air Force Base, 256th Squadron

Seoul, Korea

Mar 2022 – Dec 2023

- Led a squad of 10 airmen, overseeing daily operations, safety, and training for unit readiness.
- Managed Confidential Level II documents and coordinated flight schedules with pilots.

Mentoring Volunteer Program

Volunteer Instructor and Team Leader

Andong, Korea

Jan 2025 – Feb 2025

- Served as volunteer instructor and mentor: taught physics classes, led Sports Day activities, and provided interview coaching and STEM mentoring for high school students in a one-week program.

Skills

C++, C, Java, Python, MATLAB, RISC-V Assembly, Bluespec, PyTorch, CUDA, ROS1/2, Arduino, Fusion 360, L^AT_EX